



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Select - Digital Product Marketplace

The domain of the Project: Full Stack Web Development

Java Full Stack Development Internship

Team Mentors (and their designation):

Mr. Pruthvi Raj S

Software Engineer (**Lowe's India**)

&

MS. Chandana Tagadi

Full Stack Developer (**Atos**)

Team Members:

Mr. PINTU SWAIN
(2405292118)

Period of the project : 2 Months

December 2025 to February 2026



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Declaration

The project titled “Select - Digital Product Marketplace” has been mentored by Pruthvi Raj S & Chandana Tagadi , organised by SURE Trust, from December 2025 to February 2026, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Team Members:

Mr. Pintu swain

Signature

Mentor's Name : Mr. Pruthvi Raj S
Software Engineer
Lowe's India

Mentor's Name : Chandana Tagadi
Full Stack Developer
Atos

Prof. Radhakumari
Executive Director & Founder
SURE Trust



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Table of contents

1. Executive summary
2. Introduction
3. Project Objectives
4. Methodology & Results
5. Social / Industry relevance of the project
6. Learning & Reflection
7. Future Scope & Conclusion



Executive Summary

Overview :

"Select" is a robust, full-stack digital marketplace designed to bridge the gap between digital content creators and consumers. The primary objective of this project was to engineer a secure, scalable e-commerce platform that facilitates the instant purchase and delivery of intangible assets, such as e-books, software templates, and design resources. Unlike traditional e-commerce platforms focused on physical logistics, "Select" focuses on immediate digital fulfillment and user ease of use.

Methods and Technology :

The project was developed using a modern, industry-standard technology stack. The frontend was built with **React.js** and Tailwind CSS to ensure a responsive user interface, while the backend was powered by **Java Spring Boot** to handle complex business logic and API management. Data persistence was managed using **MySQL**, and financial transactions were secured using the **Razorpay Payment Gateway**. The development methodology emphasized component-based architecture and RESTful API communication.

Key Findings :

The project successfully demonstrated the viability of a custom-built digital distribution platform. Key technical achievements include the implementation of a secure authentication system, a persistent shopping cart, and a seamless "Buy Now" checkout flow that correctly links user credentials to transaction history. We addressed and resolved significant challenges regarding case-insensitive database queries and client-server data synchronization, ensuring a reliable user experience.

Recommendations and Value :

For stakeholders, "Select" represents a foundational model for a scalable digital economy platform. It eliminates shipping costs and inventory delays, offering a high-margin business model. We recommend future iterations focus on expanding to a multi-vendor architecture, allowing third-party creators to upload products, and developing a companion mobile application to increase user engagement. The project stands as a testament to the practical application of full-stack development skills in solving real-world e-commerce problems.



Introduction

Background and Context of the Project:

In the modern digital economy, the demand for intangible goods—such as software, e-books, digital art, and templates—has grown exponentially. Unlike traditional retail, the commerce of digital products requires no physical inventory, warehousing, or shipping logistics. However, it does require a robust technological infrastructure to handle secure file storage, instant delivery, and real-time transaction verification.

This project, titled "**Select**," was conceived to address this specific market need. It is a full-stack web application built using the powerful combination of **Java Spring Boot** for the backend and **React.js** for the frontend. This project serves as a practical implementation of modern web development practices, integrating database management, secure user authentication, and third-party payment gateways into a cohesive e-commerce solution.

Problem Statement and Goals:

Problem Statement: Many existing e-commerce platforms are designed primarily for physical goods, often resulting in a cluttered user experience for those seeking purely digital assets. Furthermore, for aspiring developers and small creators, setting up a secure platform that handles payments and instant downloads can be technically daunting and cost-prohibitive.

Goals of the Project:

- To build a specialized marketplace dedicated to digital products.
- To implement a secure, seamless "Buy Now" and "Add to Cart" workflow that feels instant to the user.
- To integrate a real-world payment gateway (**Razorpay**) to simulate a true commercial environment.
- To ensure data integrity where order history is accurately preserved and linked to specific user accounts.

Scope and Limitations:

Scope: The current scope of the "Select" marketplace includes:

- **User Management:** Secure registration and login functionality.
- **Product Catalog:** A browseable interface for various digital products with details and pricing.
- **Transaction Processing:** A fully functional shopping cart and checkout system integrated with Razorpay.
- **Order Fulfillment:** An automated system that generates download links immediately after a successful payment.
- **Order History:** A dashboard for users to view past purchases and re-download files.



Limitations:

- **Single Vendor Model:** Currently, the platform supports a single administrator/vendor uploading products. It is not yet a multi-vendor platform where users can upload their own items to sell.
- **Platform:** The application is a web-based platform; a native mobile application (Android/iOS) is not included in this phase.
- **Offline Access:** As a web application, it requires an active internet connection for all interactions.

Innovation Component :

The primary innovation in "Select" lies in its **Instant Digital Fulfillment Engine**. Unlike traditional e-commerce flows that wait for shipping status updates, "Select" automates the entire post-purchase workflow. The moment a payment signature is verified by the backend:

1. The order is logged in the database.
2. The digital asset is unlocked.
3. A secure download link is generated and presented to the user—all within milliseconds.

Additionally, the project implements a **Smart Search Optimization** on the backend. By utilizing custom JPA queries (such as `findByUserEmailIgnoreCase`), the system intelligently handles user data retrieval, preventing common database errors related to case sensitivity that often plague smaller applications.



Project Objectives

Clearly Defined Objectives and Goals

The primary objective of the "Select" project was to design, develop, and deploy a comprehensive **Full Stack Web Application** that simulates a real-world digital marketplace. The project aimed to bridge the gap between theoretical knowledge and practical industry application by utilizing the **Java Spring Boot** and **React.js** ecosystem.

Specific Goals:

1. **Full Stack Integration:** To successfully architect a seamless communication channel between a React-based frontend and a Java-based backend using RESTful APIs.
2. **Secure Authentication:** To implement a robust user authentication system that secures user data and restricts access to specific features (like checkout and order history) to logged-in users only.
3. **Real-Time Payment Processing:** To integrate the **Razorpay Payment Gateway** to handle real-currency transactions, ensuring secure validation of payment signatures on the server side.
4. **Digital Asset Management:** To create a system capable of handling digital inventory, ensuring that products (such as e-books and templates) are delivered instantly and securely upon successful payment.
5. **Data Persistence:** To design a normalized **MySQL** database schema that accurately tracks users, products, and complex order histories without data redundancy.

Expected Outcomes and Deliverables

Expected Outcomes: Upon completion of the project, the following outcomes were achieved:

- A fully functional **E-Commerce Web Application** where users can browse, add to cart, and purchase digital products.
- A **Zero-Latency Fulfillment System** where the traditional shipping process is replaced by instant download links generated post-payment.
- A **User Dashboard** ("My Orders") that persists transaction data, allowing users to track their expenditure and retrieve previously purchased files.
- **Case-Insensitive Search Functionality:** An optimized backend search feature that allows users to find products regardless of capitalization, improving the overall user experience.



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Deliverables: The project culminates in the submission of the following assets:

1. **Source Code:** The complete codebase for the Frontend (React/Vite) and Backend (Spring Boot), hosted on a shared GitHub repository.
2. **Database Schema:** The SQL scripts and Entity Relationship (ER) mappings required to reconstruct the MySQL database.
3. **Project Documentation:** A comprehensive report detailing the architecture, challenges faced (such as the "Buy Now" logic implementation), and solutions applied.
4. **User Manual:** A guide demonstrating the user flow from registration to product download.



Methodology and Results

Methods and Technology Used

The development of the "Select" marketplace followed an agile methodology, utilizing a component-based frontend and a service-oriented backend. The technology stack was chosen for its scalability, security, and industry relevance.

- **Frontend:**

- **React.js (Vite):** Utilized for building a dynamic, single-page application (SPA) that provides a smooth user experience without constant page reloads.
 - **Tailwind CSS:** Employed for rapid UI development, ensuring a responsive and modern design layout.
 - **Axios:** Used for handling asynchronous HTTP requests between the client and the server.
-

- **Backend:**

- **Java (JDK 25):** The core programming language used for business logic.
 - **Spring Boot:** The framework used to create the RESTful APIs, managing dependency injection and server configuration.
 - **Spring Data JPA / Hibernate:** Used for Object-Relational Mapping (ORM) to interact with the database using Java objects instead of raw SQL.
-



- **Database:**

-
- **MySQL:** A relational database management system used to store user credentials, product details, and order history.
-

- **Payment Gateway:**

-
- **Razorpay:** Integrated to process real-time payments and verify transactions via digital signatures.
-

Tools and Software Used

- **Integrated Development Environments (IDEs):**

-
- **IntelliJ IDEA:** Used for Backend (Java/Spring Boot) development.
 - **VS Code:** Used for Frontend (React) development.
-

- **API Testing:**

-
- **Postman:** Used to test API endpoints (GET, POST) before integrating them into the frontend.
-
-



- **Database Management:**

- **MySQL Workbench:** Used to design the schema and verify data persistence (e.g., checking orders tables).
-

- **Version Control:**

- **Git & GitHub:** Used for source code management and collaboration.
-

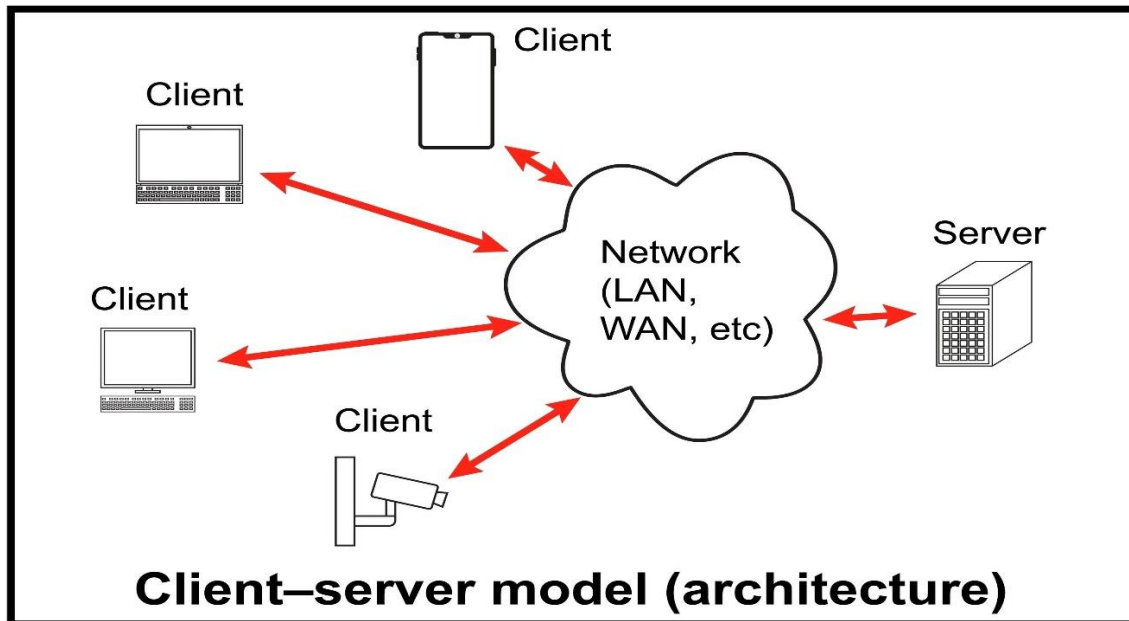
Data Collection Approach

As an e-commerce platform, the data collection strategy was **User-Centric:**

- **User Data:** Collected during registration (Name, Email, Password) and securely stored.
 - **Transactional Data:** Generated automatically when a user completes a purchase. The system captures the Amount, Product ID, Payment Signature, and User Email.
 - **Inventory Data:** Admin-generated data representing the digital products available for sale (Title, Price, Description, File Path).
-



Project Architecture



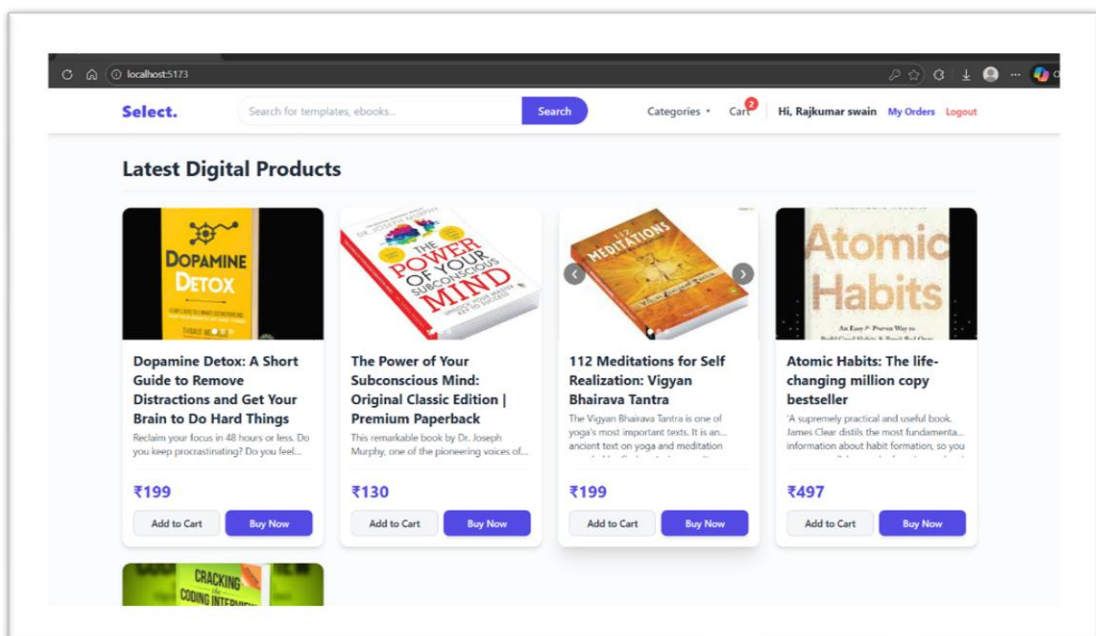
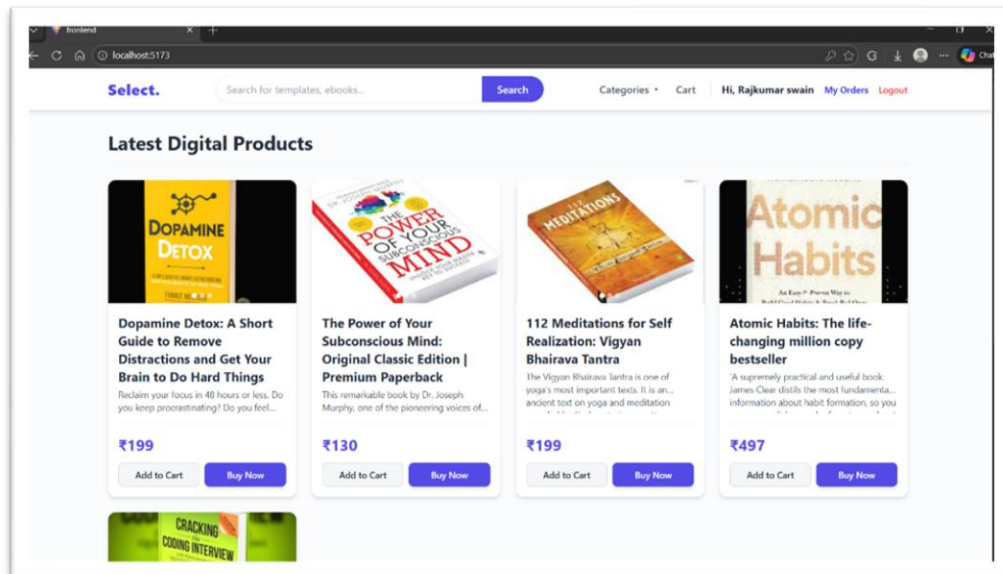
The project follows a classic **Three-Tier Architecture**:

1. **Presentation Layer (Client):** The React application runs in the user's browser. It handles all user interactions (clicks, form inputs) and displays data. When a user clicks "Buy Now," the browser sends a JSON request to the server.
 2. **Application Layer (Server):** The Spring Boot application (running on Tomcat) receives the request. The **Controller** layer maps the request to the correct function. The **Service** layer processes the business logic (e.g., verifying if the user is logged in).
 3. **Data Layer (Database):** MySQL stores the persistent data. The backend retrieves product details or saves new orders into the database tables (users, products, orders) and returns a confirmation to the client.
-



Final Project Working Screenshots

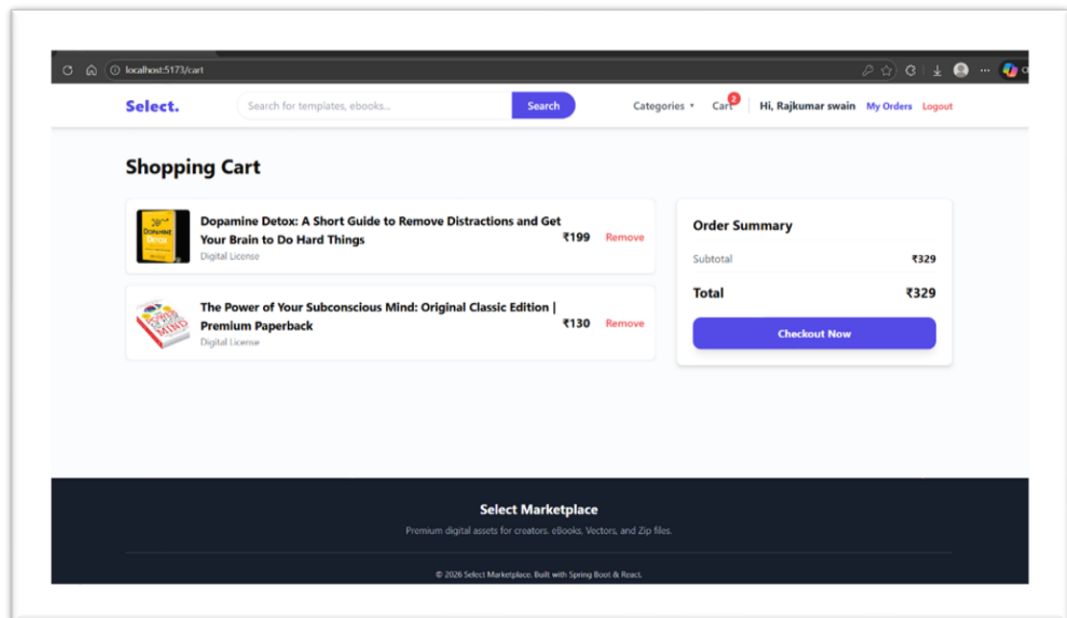
1. Homepage & Product Catalog Description: This screen displays the available digital products fetched from the database. Users can view prices, descriptions, and "Add to Cart" buttons.



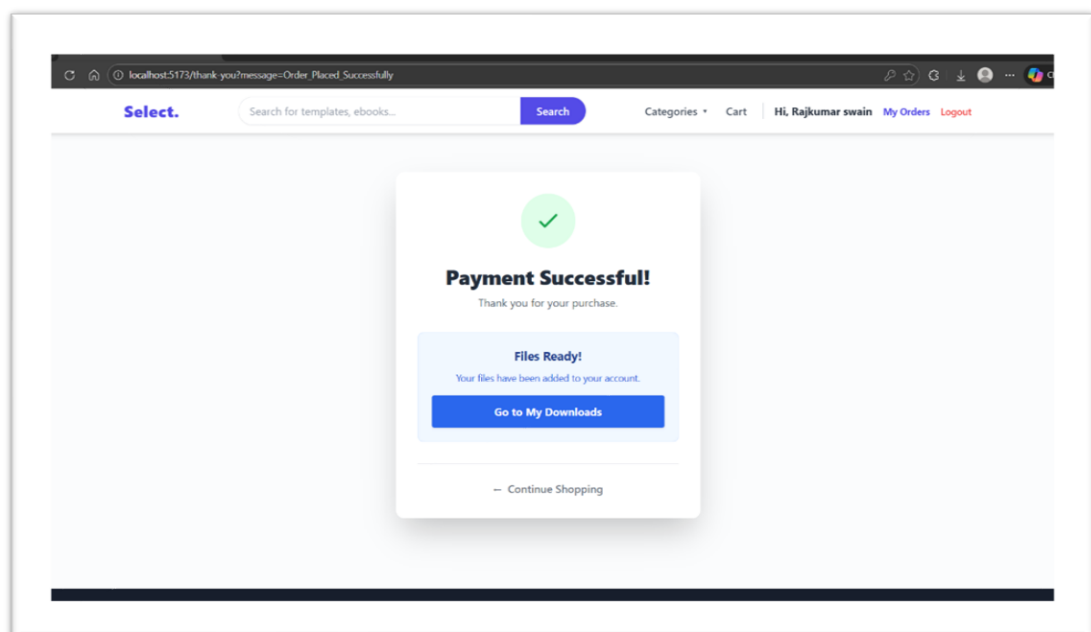


Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

2. The Shopping Cart Description: The cart page summarizes the selected items and calculates the total cost. It includes the "Checkout" button which triggers the authentication check.



3. Database & Order History Description: The MySQL Workbench view confirms that orders are being correctly saved with the User's Email and Payment Status ("PAID"), proving the end-to-end flow is successful.





Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Project GitHub Link

**[https://github.com/sure-trust/PINTU-SWAIN-g20-
java/tree/3da6ada34a8a2e494e7fcd4d3cc54501788cc41b/Final%20capstone%20project](https://github.com/sure-trust/PINTU-SWAIN-g20-java/tree/3da6ada34a8a2e494e7fcd4d3cc54501788cc41b/Final%20capstone%20project)**



Learning and Reflection

Here is the text for the **Learning and Reflection** section. You can copy and paste this directly into your report.

Learning and Reflection

New Learnings

This project served as an intensive learning ground, allowing the team to bridge the gap between theoretical concepts and practical, industry-standard application development.

Technical Learnings:

- **Full Stack Integration:** We learned how to architect a complete system where a **React.js** frontend communicates seamlessly with a **Java Spring Boot** backend. Understanding how to structure JSON requests in the frontend and map them to Java Objects (DTOs) in the backend was a critical skill acquired.
- **Payment Gateway Implementation:** Integrating **Razorpay** was a significant milestone. We learned the intricacies of handling financial transactions programmatically, including order creation, capturing payment IDs, and verifying digital signatures to prevent fraud.
- **Database Query Optimization:** We moved beyond basic SQL queries and learned how to use **JPA (Java Persistence API)** effectively. Specifically, we implemented custom query methods like `findByUserEmailIgnoreCase` to handle real-world data issues such as case sensitivity in user logins.
- **State Management:** In the frontend, we mastered the use of React Hooks (`useState`, `useEffect`, `useContext`) to manage the state of the shopping cart and user authentication across different pages.

Management & Soft Skills:

- **Debugging Strategy:** We learned that "silent failures" (where code doesn't crash but doesn't work) are often due to browser caching or logical errors. We developed a disciplined approach to debugging using browser developer tools and backend logs.
- **Version Control:** We gained practical experience using **Git and GitHub** for managing code versions, understanding the importance of keeping "secrets" (API keys) out of public repositories.



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Overall Experience

The journey of building "Select" was both challenging and deeply rewarding.

Challenges Faced: One of the most significant hurdles was the "Buy Now" button logic. We initially struggled with a bug where the user's email was returning as null in the backend, preventing orders from being saved. Through rigorous debugging—checking backend logs and inspecting frontend API calls—we identified that the browser was caching old code. Fixing this issue taught us the value of persistence and the importance of "hard refreshing" during development.

Conclusion: Overall, transforming a blank slate into a working e-commerce platform that can process real payments has given us immense confidence. It shifted our perspective from writing code snippets to building complete, resilient systems. The project not only solidified our technical foundation in Java and React but also prepared us for the collaborative and problem-solving nature of the professional software industry.



Conclusion and Future Scope

Recap of Objectives and Achievements

The primary goal of this project was to develop "**Select**," a fully functional, secure, and user-friendly digital marketplace. We set out to bridge the gap between digital content creators and consumers by eliminating physical logistics and ensuring instant delivery.

Key Achievements:

- **End-to-End E-Commerce Flow:** Successfully built a platform where users can register, browse products, add items to a cart, and complete purchases seamlessly.
- **Secure Payment Integration:** Achieved a major milestone by integrating the **Razorpay Payment Gateway**, ensuring real-time, secure transaction processing and signature verification.
- **Data Integrity:** Implemented a robust backend logic that accurately tracks user order history, resolving complex issues related to data synchronization (such as the "Buy Now" email linkage) and case-insensitive database searches.
- **Instant Fulfillment:** Realized the core value proposition of the project by automating the delivery of digital assets immediately upon payment confirmation.

Future Scope of this Project

While the current version of "Select" is a fully functional Minimum Viable Product (MVP), there is significant potential for expansion and feature enhancement in future iterations.

1. **Multi-Vendor Architecture:** Currently, the platform operates on a single-admin model. Future updates will evolve this into a true marketplace where third-party creators can register as "Vendors," upload their own digital products, and track their individual sales and revenue.
2. **Advanced Admin Dashboard:** We plan to develop a graphical Admin Interface that allows for non-technical management of inventory. This would include features for drag-and-drop product uploads, visual sales analytics (graphs/charts), and user management controls without needing to access the database directly.
3. **Automated Email Notifications:** To enhance the user experience, we aim to integrate an email service (like JavaMailSender) to send automated invoices and download links directly to the user's email inbox immediately after purchase.
4. **Mobile Application:** Given the prevalence of mobile commerce, a logical next step is to adapt the existing React code into **React Native** to deploy "Select" as a native mobile application for Android and iOS, offering push notifications for new product launches.



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

5. **Review and Rating System:** Implementing a community feedback loop where verified purchasers can leave reviews and ratings for products will build trust and assist future buyers in their decision-making process.